Proposal for Orbital 26

Team Name:
LawnMower272

Proposed Level of Achievement:

Apollo 11

Motivation

<u>Challenge in keeping up with different deadlines in University</u>

Unlike school in our earlier years, there are no teachers chasing us for deadlines in University. This experience could be new to many incoming freshmen who are not used to this high level of freedom and thus not being able to uphold the same level of responsibility when it comes to completing assignments.

Each course has vastly different requirements in terms of deadlines. Some require weekly submissions, some have occasional submissions that are spread out over random weeks. University is the perfect time to expand one's interest, leading to many freshmen participating in many different activities, whether out of interest or academically focused. This would further muddle the student's already poor grasp of which submissions are required at which timing, as they have to juggle such deadlines amidst the busy lives.

Course planning is difficult when workload demands are unclear, especially when university courses usually have inconsistent commitment requirements, with some weeks requiring my time - perhaps when a huge project is due. The current NUSMODS course planner that only displays weekly class timings and final exam dates is thus not comprehensive enough.

These issues inspired us to come up with this solution which makes timetabling for students much easier. This can smooth transition in University and reduce stress from disorganization.

Aim

We aim to develop a web application that aids in the student's planning of their time through effective tracking of deadlines and developing insight of workload of future courses through connecting with seniors/other students.

Implementing a user-friendly and customisable timetabling system can help users plan out their time well to meet specific deadlines. Such a feature can allow for users to see at a glance what are the important things to be done, leading to better time-management and less time stressing and what course they should focus on.

We are also implementing a chat system where users can share their custom schedules and connect with other students to exchange information about each course's workload. This allows students to make better informed choices, understand their options and feel more supported throughout their academic life.

Overall we aim to

1. **Improve academic life** - Through enhancing students ability to organize their time properly while meeting all necessary deadlines. Especially freshman, where our app hopes to help them decrease the stress from messy deadlines.

2. **Connect students with each other** - Features of our app allows for inter-student discussion over timetabling where students can support each other.

3. **Integrate with existing platforms** - Build upon the tools that students already have access to like NUSMODS by ensuring easy referring when needed.

User Stories

1. The first step of University is choosing the right course. As an incoming freshman who is exploring a variety of courses, I want to be able to find the course that I am interested in.

2. Additionally, apart from incoming freshmen, as a student interested in taking a module, I want to be able to see the workload of the course and the deadlines of the assignments
3. As a freshman in University, I am unsure of the demands of each of my courses and have a hard time juggling between my non-academic activities and meeting important deadlines. It would be great to see clearly when I should prioritise what.

Features

1. Timetable (core):
    - Timetable which shows when each assignment for a particular module is due
    - Ability to merge with other modules to identify weeks with higher workload
    - Colour-coded in terms of difficulty of assignment (based on past user data)
    - Ability to tick off assignments you have finished
2. Social interface (core):
    - Insightful conversations with students who have taken this module before
    - Allows incoming freshmen to connect with seniors who have taken this module, and allows them to build connections with seniors who are able to give them personal advice
3. User interface (core):
    - Personalised user profile
    - Interested Modules section
    - Ratings and comments on each module
4. AI Chatbot (extension):
    - Chatbot will be able to answer generic questions that does not require personalised advice, allowing for a quicker response time
5. Push Notifications (extension):

- Notifications will be sent as reminders to complete assignments you have yet to complete

Timeline

1. Milestone 1 - Technical proof of concept (i.e., a minimal working system with both the frontend and the backend integrated for a very simple feature)
    a. User authentication
    b. Database schema
    c. Core Timetable
2. Milestone 2 - Prototype (i.e., a working system with the core features)
    a. Ensure merging of module timetables are feasible
    b. Social interface is operational
    c. Module rating system integrated
3. Milestone 3 - Extended system (i.e., a working system with both the core + extension features)
    a. AI Chatbot successfully integrated
    b. Push notifications triggered correctly

Tech Stack

Frontend
- React.js

Backend
- Python

Database
- PostgreSQL

Qualifications

Team Member 1 : Shu Chang Yu

<u>Skills</u>
- Python
- Java
- SQL
- R

Team Member 2 : Ng Jun Kai

<u>Skills</u>
- Python
- Java
- SQL
- R
- Javascript
- HTML
- CSS

Project Experience

- BlackJack Game
- Study Pomodoro Timer

Software Engineering

- We will use Github to track errors
- We will use Github projects to map out user stories and track bugs
- We will implement unit testing for our backend code to ensure our algorithms and data transformation functions work.